

IT Capstone Project Proposal

Project Title

Uplift Gym – Online Class Booking and Sales Website

Team Members

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- Jacob Castro
- Antonio George
- Bianca Guimaraes Barros Leal Santos

What's the Problem?

Many small gyms still organize classes in a very manual way. Customers usually need to call, text, or send messages just to ask about schedules or available spots. This takes time and often causes confusion.

Some common problems are:

- Classes getting overbooked
- Customers do not know how many spots are left
- No easy way to buy classes online
- Staff repeating the same information all the time
- Poor organization of class schedules

The gym needs a simple website where customers can see classes, book a spot, and buy class packages without any trouble.

Our Idea (The Solution)

Our group will build Uplift Gym, a website for a small gym that offers different fitness classes.

With this website, customers will be able to:

- See all available classes
- Check class schedules
- See how many spots are left
- Book a class online

- Buy class packages

Each class will have a limit of 20 people. When someone books a class, the system will automatically update the number of available spots.

What the Website Will Have

Class Booking

- At least 10 different classes
- Each class has a 20-person limit
- Shows real-time availability (example: “6 spots left”)
- Stops bookings when the class is full

Online Sales

- Customers can buy:
 - Single classes
 - Class packages
- Simple shopping cart
- Simulated checkout

Design

- Built with HTML, CSS, and JavaScript
- Clean and easy-to-use design
- Works on desktop and mobile

Database

The database will store:

- Class names and descriptions
- Class schedules (date and time)
- Total and remaining spots
- Bookings
- Products for sale

This allows the system to update class availability automatically.

Tools We Will Use

- HTML / CSS / JavaScript – build the website
- Database – store classes and bookings
- GitHub – version control and teamwork
- GitHub Pages – host the website
- Microsoft Project – plan tasks and timelines

How This Meets the Course Requirements

- Solves a real business problem
- Includes online sales
- Uses web technologies
- Uses a database for real-time updates
- Uses GitHub for version control
- Uses MS Project for planning
- Shows teamwork between group members

Team Roles

Person 1 – Back-End Developer (Muhammad Fotih Abdullaev)

- Create and manage the database
- Handle class bookings and capacity limits
- Make sure classes cannot be overbooked

Person 2 – Back-End Developer (Jacob Castro)

- Manage class schedules and class types
- Help with booking logic and data validation
- Support testing and fixing database issues

Person 3 – Front-End Developer (Bianca Guimaraes Barros Leal Santos)

- Build website pages (classes, booking, cart)

- Style the site using HTML and CSS
- Make sure the website works well on desktop and mobile

Person 4 – Front-End Developer (Antonio George)

- Add JavaScript for booking, forms, and alerts
- Connect front-end pages to the database
- Improve user interaction and feedback

Even though each person has a main role, we will all help each other and step in when needed. The goal is to work as a team and make sure the project gets done the right way.

Example Class List (10 Classes)

1. Yoga
2. HIIT
3. Pilates
4. Spin
5. Strength Training
6. Zumba
7. Functional Training
8. Boxing Fitness
9. Mobility & Stretch
10. Meditation

Final Thoughts

Uplift Gym is a simple and realistic capstone project. It helps gyms stay organized and makes it easier for customers to book classes. This project shows the skills we've learned in our IT and Programming courses and helps prepare us for the rest of the semester.